

Chapter 4

Comparisons of Several Multivariate Means

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- Repeated Measures within Subjects Across Treatments (Randomized Block Design)
- Comparing 2 Treatments/Populations with Independent Samples
- 1-Way Analysis of Variance with g Treatments/Groups
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Paired Comparisons: Univariate Case

Single Variable Observed within subjects/units in 2 conditions (e.g. Before/After)

$$\begin{bmatrix} X_{j1} \\ X_{j2} \end{bmatrix} \stackrel{i.i.d}{\sim} \mathcal{N} \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_{11} & \rho\sqrt{\sigma_{11}\sigma_{22}} \\ \rho\sqrt{\sigma_{11}\sigma_{22}} & \sigma_{22} \end{bmatrix} \right)$$

Let $D_j = X_{j1} - X_{j2} \quad j = 1, \dots, n$

$$\Rightarrow D_j \stackrel{i.i.d}{\sim} \mathcal{N}(\delta, \sigma_d^2) \quad \text{where} \quad \delta = \mu_1 - \mu_2 \quad \text{and} \quad \sigma_d^2 = \sigma_{11} + \sigma_{22} - 2\rho\sqrt{\sigma_{11}\sigma_{22}}$$

The variable

$$T = \frac{\bar{D} - \delta}{S_d / \sqrt{n}} \sim t_{n-1}$$

$$\text{where} \quad \bar{D} = \frac{1}{n} \sum_{j=1}^n D_j \quad \text{and} \quad S_d^2 = \frac{1}{n-1} \sum_{j=1}^n (D_j - \bar{D})^2$$

Paired Comparisons: Univariate Case

Sample Measurements: $(x_{11}, x_{12}), \dots, (x_{n1}, x_{n2})$

$$\rightarrow d_j = x_{j1} - x_{j2} \quad , \quad \bar{d} = \frac{1}{n} \sum_{j=1}^n d_j, \quad s_d^2 = \frac{1}{n-1} \sum_{j=1}^n (d_j - \bar{d})^2$$

An α -level test of

$$H_0 : \delta = 0 \quad (\text{zero mean difference for treatments}) \quad \text{vs} \quad H_A : \delta \neq 0$$

$$TS : t_{obs} = \frac{\bar{d} - 0}{s_d / \sqrt{n}} \quad RR : |t_{obs}| \geq t_{n-1} \left(\frac{\alpha}{2} \right) \quad p_{value} = 2P(t_{n-1} \geq |t_{obs}|)$$

$$(1 - \alpha) 100\% \text{ Confidence Interval for } \delta : \quad \bar{d} \pm t_{n-1} \left(\frac{\alpha}{2} \right) \left[\frac{s_d}{\sqrt{n}} \right].$$

Paired Comparisons: Multivariate Case - I

Multivariate Case: n Subjects/Units, 2 Treatments /Conditions, p Outcome Measures

Subject j : Treatment 1: $X_{1j1}, \dots, X_{1jp}$; Treatment 2: $X_{2j1}, \dots, X_{2jp}$

$$\mathbf{D}_j = \begin{bmatrix} D_{j1} = X_{1j1} - X_{2j1} \\ \vdots \\ D_{jp} = X_{1jp} - X_{2jp} \end{bmatrix} \quad E\{\mathbf{D}_j\} = \boldsymbol{\delta} = \begin{bmatrix} \delta_1 \\ \vdots \\ \delta_p \end{bmatrix} \quad V\{\mathbf{D}_j\} = \boldsymbol{\Sigma}_d$$

$$\bar{\mathbf{D}} = \frac{1}{n} \sum_{j=1}^n \mathbf{D}_j \quad \mathbf{S}_d = \frac{1}{n-1} \sum_{j=1}^n (\mathbf{D}_j - \bar{\mathbf{D}}) (\mathbf{D}_j - \bar{\mathbf{D}})' \quad j = 1, \dots, n$$

Result 1:

Let $D_j \stackrel{i.i.d}{\sim} \mathcal{N}_p(\boldsymbol{\delta}, \boldsymbol{\Sigma}_d)$ then $T^2 = n(\bar{\mathbf{D}} - \boldsymbol{\delta})' \mathbf{S}_d^{-1} (\bar{\mathbf{D}} - \boldsymbol{\delta})$

$$\text{and} \quad \frac{n-p}{p(n-1)} T^2 \sim \mathcal{F}_{p, n-p}$$

If n and n - p are both large $T^2 \sim \chi_p^2$, regardless of the distribution of the differences.

Paired Comparisons: Multivariate Case - II

Multivariate Case:

$$\text{Defining: } \mathbf{D}^* = \begin{bmatrix} \mathbf{D}_1' \\ \vdots \\ \mathbf{D}_n' \end{bmatrix} \Rightarrow \mathbf{S}_d = \frac{1}{n-1} \mathbf{D}^{*'} \left[\mathbf{I}_n - \frac{1}{n} \mathbf{J}_n \right] \mathbf{D}^*$$

$$\text{Observed differences: } \mathbf{d}_j' = [d_{j1} \quad \cdots \quad d_{jp}]$$

$$\bar{\mathbf{d}} = \frac{1}{n} \sum_{j=1}^n \mathbf{d}_j \quad \mathbf{S}_d = \frac{1}{n-1} \sum_{j=1}^n (\mathbf{d}_j - \bar{\mathbf{d}}) (\mathbf{d}_j - \bar{\mathbf{d}})' \quad T_{obs}^2 = n \bar{\mathbf{d}}' \mathbf{S}_d^{-1} \bar{\mathbf{d}}$$

$$\text{Hypothesis: } H_0 : \boldsymbol{\delta} = \mathbf{0} \quad \text{vs} \quad H_A : \boldsymbol{\delta} \neq \mathbf{0}$$

$$TS : \frac{n-p}{p(n-1)} T^2 \quad RR : \frac{n-p}{p(n-1)} T_{obs}^2 \geq F_{p, n-p}(\alpha)$$

$$p_{\text{value}} = P \left(F_{p, n-p} \geq \frac{n-p}{p(n-1)} T^2 \right)$$

Paired Comparisons: Multivariate Case - III

$(1 - \alpha) 100\%$ **Confidence Region for δ** :

$$\text{Values of } \delta \text{ s.t. } (\bar{\mathbf{d}} - \delta)' \mathbf{S}_d^{-1} (\bar{\mathbf{d}} - \delta) \leq \left[\frac{p(n-1)}{n(n-p)} \right] F_{p,n-p}(\alpha)$$

Simultaneous $(1 - \alpha) 100\%$ Confidence Intervals for Individual Differences δ_i , $i = 1, \dots, p$

$$\bar{d}_i \pm \sqrt{\left[\frac{p(n-1)}{n(n-p)} \right] F_{p,n-p}(\alpha)} \sqrt{\frac{s_{di}^2}{n}}$$

For large n , $n - p$ without assumption of normality on differences:

$\left[\frac{p(n-1)}{n(n-p)} \right] F_{p,n-p}(\alpha)$ can be replaced by $\chi_p^2(\alpha)$ for approximate CI

Bonferroni Simultaneous $(1 - \alpha) 100\%$ CI's for $\delta_1, \dots, \delta_p$:

$$\bar{d}_i \pm t_{n-1} \left(\frac{\alpha}{2p} \right) \sqrt{\frac{s_{di}^2}{n}}$$

Paired Comparisons: Multivariate Case - IV

Computation from full sample:

$$\bar{\mathbf{x}} = \begin{bmatrix} \bar{x}_{11} \\ \vdots \\ \bar{x}_{1p} \\ \bar{x}_{21} \\ \vdots \\ \bar{x}_{2p} \end{bmatrix} = \begin{bmatrix} \bar{\mathbf{x}}_1 \\ \bar{\mathbf{x}}_2 \end{bmatrix}$$

$$\mathbf{S} = \begin{bmatrix} \mathbf{S}_{11} & \mathbf{S}_{12} \\ \mathbf{S}_{12} & \mathbf{S}_{22} \end{bmatrix} \quad \mathbf{S}_{ik} = \frac{1}{n-1} \sum_{j=1}^n (\mathbf{x}_{ji} - \bar{\mathbf{x}}_i) (\mathbf{x}_{jk} - \bar{\mathbf{x}}_k)'$$

$$\mathbf{C} = \begin{bmatrix} \mathbf{I}_p & -\mathbf{I}_p \end{bmatrix} \quad \mathbf{C}\bar{\mathbf{x}} = \bar{\mathbf{d}} \quad \mathbf{CSC}' = \mathbf{S}_d$$

$$\Rightarrow T^2 = n\bar{\mathbf{x}}' \mathbf{C}' (\mathbf{CSC}')^{-1} \mathbf{C}\bar{\mathbf{x}}$$

Repeated Measures for Comparing q Treatments Within Subjects - I

n Subjects/Units, 1 Outcome Variable, q Treatments/Conditions

Multivariate Approach To Randomized Block Design

$$\mathbf{X}_j = \begin{bmatrix} X_{j1} \\ \vdots \\ X_{jq} \end{bmatrix} \quad j = 1, \dots, n \quad E\{\mathbf{X}_j\} = \boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_q \end{bmatrix} \quad V\{\mathbf{X}_j\} = \boldsymbol{\Sigma}$$

$$\bar{\mathbf{X}} = \frac{1}{n} \sum_{j=1}^n \mathbf{X}_j \quad E\{\bar{\mathbf{X}}\} = \boldsymbol{\mu} \quad V\{\bar{\mathbf{X}}\} = \frac{1}{n} \boldsymbol{\Sigma}$$

Repeated Measures for Comparing q Treatments Within Subjects - II

Contrast Vectors/Matrices (2 Examples given here):

$$1) \begin{bmatrix} \mu_1 - \mu_2 \\ \mu_1 - \mu_3 \\ \vdots \\ \mu_1 - \mu_q \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 & \cdots & 0 \\ 1 & 0 & -1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & -1 \end{bmatrix} \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_q \end{bmatrix} = \mathbf{C}_1 \boldsymbol{\mu}$$

$$2) \begin{bmatrix} \mu_2 - \mu_1 \\ \mu_3 - \mu_2 \\ \vdots \\ \mu_q - \mu_{q-1} \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & 1 \end{bmatrix} \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_q \end{bmatrix} = \mathbf{C}_2 \boldsymbol{\mu}$$

$$\mu_1 = \mu_2 = \dots = \mu_q \Rightarrow \mathbf{C}_1 \boldsymbol{\mu} = \mathbf{C}_2 \boldsymbol{\mu} = \mathbf{0}$$

$$E \{ \mathbf{C} \bar{\mathbf{X}} \} = \mathbf{C} \boldsymbol{\mu} \quad V \{ \mathbf{C} \bar{\mathbf{X}} \} = \frac{1}{n} \mathbf{C} \boldsymbol{\Sigma} \mathbf{C}'$$

Repeated Measures for Comparing q Treatments Within Subjects - III

Observed Measurements: $j = 1, \dots, n$

$$\mathbf{x}_j = \begin{bmatrix} x_{j1} \\ \vdots \\ x_{jq} \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} \mathbf{x}_1' \\ \vdots \\ \mathbf{x}_n' \end{bmatrix} \quad \rightarrow \quad \bar{\mathbf{x}} = \frac{1}{n} \sum_{j=1}^n \mathbf{x}_j = \mathbf{X}' \frac{1}{n} \mathbf{J}_n$$

$$\mathbf{S} = \frac{1}{n-1} \sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{x}}) (\mathbf{x}_j - \bar{\mathbf{x}})' = \frac{1}{n-1} \mathbf{X}' \left[\mathbf{I}_n - \frac{1}{n} \mathbf{J}_n \right] \mathbf{X}$$

 $T^2 = n(\mathbf{C}\bar{\mathbf{x}})' (\mathbf{CSC}')^{-1} \mathbf{C}\bar{\mathbf{x}}$ which is invariant to which Contrast Matrix is used

$$H_0 : \mathbf{C}\mu = \mathbf{0} \quad \text{vs} \quad H_A : \mathbf{C}\mu \neq \mathbf{0}$$

$$TS : \left[\frac{n - (q-1)}{(n-1)(q-1)} \right] T^2 \quad RR : \left[\frac{n - (q-1)}{(n-1)(q-1)} \right] T^2 \geq F_{q-1, n-(q-1)}(\alpha)$$

$$p_{value} = P \left(F_{q-1, n-(q-1)} \geq \left[\frac{n - (q-1)}{(n-1)(q-1)} \right] T^2 \right)$$

Repeated Measures for Comparing q Treatments Within Subjects - IV

$(1 - \alpha) 100\%$ **Confidence Region for $C\mu$:** $C\mu$ s.t.

$$\left[\frac{n - (q - 1)}{(n - 1)(q - 1)} \right] n (C\bar{x} - C\mu)' (CSC')^{-1} (C\bar{x} - C\mu) \leq F_{q-1, n-(q-1)}(\alpha)$$

Simultaneous $(1 - \alpha) 100\%$ CIs for single contrasts: $c'\mu$

$$c'\bar{x} \pm \sqrt{\frac{(n - 1)(q - 1)}{n - (q - 1)} F_{q-1, n-(q-1)}(\alpha)} \sqrt{\frac{c'Sc}{n}}$$

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Comparing 2 Populations/Treatments – Independent Samples - I

$$\text{Sample 1: } \mathbf{X}_{11}, \dots, \mathbf{X}_{1n_1} \quad \mathbf{X}_{1j} = \begin{bmatrix} X_{1j1} \\ \vdots \\ X_{1jp} \end{bmatrix} \quad E\{\mathbf{X}_{1j}\} = \boldsymbol{\mu}_1 \quad V\{\mathbf{X}_{1j}\} = \boldsymbol{\Sigma}_1$$

$$\bar{\mathbf{X}}_1 = \frac{1}{n_1} \sum_{j=1}^{n_1} \mathbf{X}_{1j} \quad \mathbf{S}_1 = \frac{1}{n_1 - 1} \sum_{j=1}^{n_1} (\mathbf{X}_{1j} - \bar{\mathbf{X}}_1) (\mathbf{X}_{1j} - \bar{\mathbf{X}}_1)'$$

$$\text{Sample 2: } \mathbf{X}_{21}, \dots, \mathbf{X}_{2n_2} \quad \mathbf{X}_{2j} = \begin{bmatrix} X_{2j1} \\ \vdots \\ X_{2jp} \end{bmatrix} \quad E\{\mathbf{X}_{2j}\} = \boldsymbol{\mu}_2 \quad V\{\mathbf{X}_{2j}\} = \boldsymbol{\Sigma}_2$$

$$\bar{\mathbf{X}}_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} \mathbf{X}_{2j} \quad \mathbf{S}_2 = \frac{1}{n_2 - 1} \sum_{j=1}^{n_2} (\mathbf{X}_{2j} - \bar{\mathbf{X}}_2) (\mathbf{X}_{2j} - \bar{\mathbf{X}}_2)'$$

$$\{\mathbf{X}_{1j}\} \perp \{\mathbf{X}_{2j}\}.$$

For small sample, additional necessary assumptions for inference:

$$1) \quad \mathbf{X}_{ij} \stackrel{i.i.d}{\sim} \mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i) \quad 2) \quad \boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2$$

Comparing 2 Populations/Treatments – Independent Samples – II

Pooled Estimated Variance - Covariance Matrix:

$$\mathbf{S}_{\text{Pooled}} = \frac{(n_1-1)\mathbf{S}_1 + (n_2-1)\mathbf{S}_2}{n_1 + n_2 - 2}$$

$$= \frac{\sum_{j=1}^{n_1} (\mathbf{x}_{1j} - \bar{\mathbf{X}}_1) (\mathbf{x}_{1j} - \bar{\mathbf{X}}_1)' + \sum_{j=1}^{n_2} (\mathbf{x}_{2j} - \bar{\mathbf{X}}_2) (\mathbf{x}_{2j} - \bar{\mathbf{X}}_2)'}{n_1 + n_2 - 2}$$

$$E \{ \bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 \} = \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$$

$$V \{ \bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 \} = V \{ \bar{\mathbf{X}}_1 \} + V \{ \bar{\mathbf{X}}_2 \} - 2\text{COV} \{ \bar{\mathbf{X}}_1, \bar{\mathbf{X}}_2 \} = \frac{1}{n_1} \boldsymbol{\Sigma}_1 + \frac{1}{n_2} \boldsymbol{\Sigma}_2 - 2(\mathbf{0})$$

$$= \left(\frac{1}{n_1} + \frac{1}{n_2} \right) \boldsymbol{\Sigma}$$

$$\hat{V} \{ \bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 \} = \left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_{\text{Pooled}}$$

Comparing 2 Populations/Treatments – Independent Samples – III

Observed Samples: $x_{i1}, \dots, x_{in_i}$ $i = 1, 2 \rightarrow \bar{\mathbf{x}}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \mathbf{x}_{ij}$

$$\mathbf{S}_i = \frac{1}{n_i - 1} \sum_{j=1}^{n_i} (\mathbf{x}_{ij} - \bar{\mathbf{x}}_i) (\mathbf{x}_{ij} - \bar{\mathbf{x}}_i)' \quad \mathbf{S}_{\text{Pooled}} = \frac{(n_1 - 1) \mathbf{S}_1 + (n_2 - 1) \mathbf{S}_2}{n_1 + n_2 - 2}$$

Testing whether $\mu_1 - \mu_2 = \delta_0$:

$$T^2 = ((\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - \delta_0)' \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_{\text{Pooled}} \right]^{-1} ((\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - \delta_0)$$

$$H_0 : \mu_1 - \mu_2 = \delta_0 \quad \text{vs} \quad H_A : \mu_1 - \mu_2 \neq \delta_0$$

$$TS : \left[\frac{n_1 + n_2 - p - 1}{p(n_1 + n_2 - 2)} \right] T^2 \quad RR : \left[\frac{n_1 + n_2 - p - 1}{p(n_1 + n_2 - 2)} \right] T^2 \geq F_{p, n_1 + n_2 - p - 1}(\alpha)$$

$$p_{\text{value}} = P \left(F_{p, n_1 + n_2 - p - 1} \geq \left[\frac{n_1 + n_2 - p - 1}{p(n_1 + n_2 - 2)} \right] T^2 \right)$$

Comparing 2 Populations/Treatments – Independent Samples – IV

$(1 - \alpha)$ 100% Confidence Ellipsoid for δ : All δ s.t. :

$$((\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - \delta)' \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_{\text{Pooled}} \right]^{-1} ((\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - \delta) \leq \left[\frac{p(n_1 + n_2 - 2)}{n_1 + n_2 - p - 1} \right] F_{p, n_1 + n_2 - p - 1}(\alpha) = c^2$$

The Confidence Ellipse extends $\sqrt{\lambda_i} \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2} \right) c^2}$ in the direction of i^{th} eigenvector of $\mathbf{S}_{\text{Pooled}}$.

Simultaneous $(1 - \alpha)$ 100% CI's for $\mathbf{a}'(\mu_1 - \mu_2)$:

$$\mathbf{a}'(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) \pm \sqrt{c^2} \sqrt{\mathbf{a}' \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{S}_{\text{Pooled}} \right] \mathbf{a}}$$

Linear combination with largest population difference:

$$\hat{\mathbf{a}}'(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) \propto \mathbf{S}_{\text{Pooled}}^{-1}(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)$$

Bonferroni $(1 - \alpha)$ 100% CI's for p Mean Differences:

$$(\bar{x}_{1i} - \bar{x}_{2i}) \pm t_{n_1 + n_2 - 2} \left(\frac{\alpha}{2p} \right) \sqrt{s_{ii, \text{Pooled}} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

Comparing 2 Populations/Treatments – Independent Samples - V

When $\Sigma_1 \neq \Sigma_2$, Case 1: Large $n_1 - p$, $n_2 - p$

Approximate $(1 - \alpha)$ 100% Confidence Ellipsoid for $\delta = \mu_1 - \mu_2$: δ s.t.;

$$((\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - \delta)' \left[\frac{1}{n_1} \mathbf{S}_1 + \frac{1}{n_2} \mathbf{S}_2 \right]^{-1} ((\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - \delta) \leq \chi_p^2(\alpha)$$

Approx. to Distribution of T^2 for Normal Populations with small n_1 , n_2

$$T^2 = ((\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - \delta)' \left[\frac{1}{n_1} \mathbf{S}_1 + \frac{1}{n_2} \mathbf{S}_2 \right]^{-1} ((\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - \delta)$$

$$H_0 : \delta = \mu_1 - \mu_2 = \mathbf{0} \quad \text{vs} \quad H_A : \delta = \mu_1 - \mu_2 \neq \mathbf{0}$$

$$TS : T^2 = ((\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - \mathbf{0})' \left[\frac{1}{n_1} \mathbf{S}_1 + \frac{1}{n_2} \mathbf{S}_2 \right]^{-1} ((\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2) - \mathbf{0})$$

$$RR : T^2 \geq \frac{\nu p}{\nu - p + 1} F_{p, \nu - p + 1}(\alpha)$$

$$\text{with } \nu = \frac{p + p^2}{\sum_{i=1}^2 \frac{1}{n_i} \left[\text{tr} \left(\left(\frac{1}{n_i} \mathbf{S}_i \left(\frac{1}{n_1} \mathbf{S}_1 + \frac{1}{n_2} \mathbf{S}_2 \right)^{-1} \right)^2 \right) + \left(\text{tr} \left(\frac{1}{n_i} \mathbf{S}_i \left(\frac{1}{n_1} \mathbf{S}_1 + \frac{1}{n_2} \mathbf{S}_2 \right)^{-1} \right) \right)^2 \right]}.$$

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1-Way ANOVA – Univariate Case I

g Treatments/Populations n_i observations for Treatment i $N = \sum_{l=1}^g n_l$

$$X_{lj} = \mu_l + e_{lj} = \mu + \tau_l + \epsilon_{lj} \quad l = 1, \dots, g; j = 1, \dots, n_l$$

$$\epsilon_{lj} \stackrel{i.i.d}{\sim} \mathcal{N}(0, \sigma^2) \quad \text{and} \quad \sum_{l=1}^g n_l \tau_l = 0$$

Observed data: $x_{11}, \dots, x_{1n_1}, \dots, x_{g1}, \dots, x_{gn_g}$

$$\rightarrow \bar{x}_l = \frac{1}{n_l} \sum_{j=1}^{n_l} x_{lj} \quad \bar{x} = \frac{1}{N} \sum_{l=1}^g \sum_{j=1}^{n_l} x_{lj} = \frac{1}{N} \sum_{l=1}^g n_l \bar{x}_l$$

Sums of Squares:

$$\text{Total: } TSS = \sum_{l=1}^g \sum_{j=1}^{n_l} (x_{lj} - \bar{x})^2 \quad \text{Error: } SSE = \sum_{l=1}^g \sum_{j=1}^{n_l} (x_{lj} - \bar{x}_l)^2$$

$$\text{Treatment: } SST = \sum_{l=1}^g \sum_{j=1}^{n_l} (\bar{x}_l - \bar{x})^2 = \sum_{l=1}^g n_l (\bar{x}_l - \bar{x})^2$$

1-Way ANOVA – Univariate Case II

ANOVA				
Source	df	SS	MS	F
Treatments	g-1	SST	$MST=SST/(g-1)$	$F=MST/MSE$
Error	N-g	SSE	$MSE=SSE/(N-g)$	
Total	N-1	TSS		

$$H_0 : \mu_1 = \dots = \mu_g \quad (\tau_1 = \dots = \tau_g = 0)$$

$$TS : F_{obs} = \frac{MST}{MSE} \quad RR : F_{obs} \geq F_{g-1, N-g}(\alpha)$$

$$p_{value} = P(F_{g-1, N-g} \geq F_{obs})$$

1-Way ANOVA – Multivariate Case - I

$$\mathbf{x}_{lj} = \begin{bmatrix} X_{lj1} \\ \vdots \\ X_{lj p} \end{bmatrix} = \boldsymbol{\mu} + \boldsymbol{\tau}_l + \boldsymbol{\epsilon}_{lj} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_p \end{bmatrix} + \begin{bmatrix} \tau_{l1} \\ \vdots \\ \tau_{lp} \end{bmatrix} + \begin{bmatrix} \epsilon_{lj1} \\ \vdots \\ \epsilon_{lj p} \end{bmatrix} \quad \mathbf{e}_{lj} \stackrel{i.i.d}{\sim} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$$

Observed Data:

$$\mathbf{x}_{lj} - \bar{\mathbf{x}} = (\bar{\mathbf{x}}_l - \bar{\mathbf{x}}) + (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l) \quad \bar{\mathbf{x}}_l = \frac{1}{n_l} \sum_{j=1}^{n_l} \mathbf{x}_{lj} \quad \bar{\mathbf{x}} = \frac{1}{N} \sum_{l=1}^g \sum_{j=1}^{n_l} \mathbf{x}_{lj} = \frac{1}{N} \sum_{l=1}^g n_l \bar{\mathbf{x}}_l$$

$$\Rightarrow (\mathbf{x}_{lj} - \bar{\mathbf{x}}) (\mathbf{x}_{lj} - \bar{\mathbf{x}})' = (\bar{\mathbf{x}}_l - \bar{\mathbf{x}}) (\bar{\mathbf{x}}_l - \bar{\mathbf{x}})' + (\bar{\mathbf{x}}_l - \bar{\mathbf{x}}) (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l)' + (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l) (\bar{\mathbf{x}}_l - \bar{\mathbf{x}})' + (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l) (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l)'$$

$$\text{Note: } \sum_{l=1}^g \sum_{j=1}^{n_l} (\bar{\mathbf{x}}_l - \bar{\mathbf{x}}) (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l)' = \mathbf{0}$$

$$\Rightarrow \sum_{l=1}^g \sum_{j=1}^{n_l} (\mathbf{x}_{lj} - \bar{\mathbf{x}}) (\mathbf{x}_{lj} - \bar{\mathbf{x}})' = \sum_{l=1}^g n_l (\bar{\mathbf{x}}_l - \bar{\mathbf{x}}) (\bar{\mathbf{x}}_l - \bar{\mathbf{x}})' + \sum_{l=1}^g \sum_{j=1}^{n_l} (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l) (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l)'$$

Total Corrected SSCP Matrix (**B** + **W**) = Between Treatment SSCP Matrix (**B**) +
 Within Treatment SSCP Matrix (**W**)

1-Way ANOVA – Multivariate Case - II

Within Treatment SSCP: $\mathbf{W} = \sum_{l=1}^g \sum_{j=1}^{n_l} (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l) (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l)' = \sum_{l=1}^g (n_l - 1) \mathbf{S}_l$

where $\mathbf{S}_l$ is the estimated variance - covariance for the l^{th} treatment/group.

Multivariate Analysis of Variance (MANOVA) Table:

Source of Variation	df	SSCP
Treatment	$g - 1$	$\mathbf{B} = \sum_{l=1}^g n_l (\bar{\mathbf{x}}_l - \bar{\mathbf{x}}) (\bar{\mathbf{x}}_l - \bar{\mathbf{x}})'$
Error (Residual)	$N - g$	$\mathbf{W} = \sum_{l=1}^g \sum_{j=1}^{n_l} (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l) (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l)'$
Total (Corrected)	$N - 1$	$\mathbf{B} + \mathbf{W} = \sum_{l=1}^g \sum_{j=1}^{n_l} (\mathbf{x}_{lj} - \bar{\mathbf{x}}) (\mathbf{x}_{lj} - \bar{\mathbf{x}})'$

Test for Treatment Effects with Wilks' Lambda:

$$H_0 : \tau_1 = \dots = \tau_g = \mathbf{0}$$

$$TS : \Lambda^* = \frac{|\mathbf{W}|}{|\mathbf{B} + \mathbf{W}|} \quad RR : \Lambda^* \leq c^* \text{ (Depending on } p \text{ and } g)$$

1-Way ANOVA – Multivariate Case - III

Sampling Distribution of Wilks' Lambda for Multivariate Normal Data and Various p and g Values

$p =$ # Variables	$g =$ # Treatments	Sampling Distribution
$p = 1$	$g \geq 2$	$[(N-g)/(g-1)][(1-L^*)/L^*] \sim F_{g-1, N-g}$
$p = 2$	$g \geq 2$	$[(N-g-1)/(g-1)][(1-(L^*)^{1/2})/(L^*)^{1/2}] \sim F_{2(g-1), 2(N-g-1)}$
$p \geq 1$	$g = 2$	$[(N-p)/(p-1)][(1-L^*)/L^*] \sim F_{p, N-p-1}$
$p \geq 1$	$g = 3$	$[(N-p-2)/p][(1-(L^*)^{1/2})/(L^*)^{1/2}] \sim F_{2p, 2(N-p-2)}$

Large - Sample Results with $H_0 : \tau_1 = \dots = \tau_g = \mathbf{0}$ and large N :

$$-\left(N - 1 - \frac{(p+g)}{2}\right) \ln \Lambda^* = -\left(N - 1 - \frac{(p+g)}{2}\right) \ln \left(\frac{|\mathbf{W}|}{|\mathbf{B} + \mathbf{W}|} \right) \sim \chi^2_{p(g-1)}$$

$$RR : -\left(N - 1 - \frac{(p+g)}{2}\right) \ln \left(\frac{|\mathbf{W}|}{|\mathbf{B} + \mathbf{W}|} \right) \geq \chi^2_{p(g-1)}(\alpha)$$

1-Way ANOVA – Multivariate Case - IV

Comparisons among Treatment Means within Response Variables:

$$\bar{X}_{li} = \frac{1}{n_l} \sum_{j=1}^{n_l} X_{lji} \quad V\{\bar{X}_{li}\} = \frac{\sigma_{ii}}{n_l} \quad V\{\bar{X}_{li} - \bar{X}_{l'i}\} = \sigma_{ii} \left(\frac{1}{n_l} + \frac{1}{n_{l'}} \right)$$

$$\mathbf{S} = \frac{1}{N-g} \mathbf{W} \Rightarrow \hat{V}\{\bar{X}_{li} - \bar{X}_{l'i}\} = \frac{w_{ii}}{N-g} \left(\frac{1}{n_l} + \frac{1}{n_{l'}} \right)$$

with $w_{ii} \equiv i^{th}$ diagonal element of $\mathbf{W}$; $l = 1, \dots, g$; $i = 1, \dots, p$.

Observed Data $\rightarrow \bar{X}_{li} = \frac{1}{n_l} \sum_{j=1}^{n_l} x_{lji}$

Bonferroni Simultaneous Confidence Intervals (All pairs of Treatments within all Variables):

$$(\bar{X}_{li} - \bar{X}_{l'i}) \pm t_{N-g} \left(\frac{\alpha}{2m} \right) \sqrt{\frac{w_{ii}}{N-g} \left(\frac{1}{n_l} + \frac{1}{n_{l'}} \right)}$$

with $m = p \binom{g}{2} = \frac{pg(g-1)}{2}$.

1-Way ANOVA – Multivariate Case - V

4 Approximate F – Tests Based on Eigenvalues of $\mathbf{W}^{-1}\mathbf{B}$:

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p > 0$$

$$\text{Wilks' } \Lambda : \Lambda = \prod_{i=1}^p \frac{1}{1 + \lambda_i} = \frac{|\mathbf{W}|}{|\mathbf{B} + \mathbf{W}|}$$

$$\text{Pillai's Trace: } V = \sum_{i=1}^p \frac{\lambda_i}{1 + \lambda_i} = \text{trace} \left(\mathbf{B}(\mathbf{B} + \mathbf{W})^{-1} \right)$$

$$\text{Hotelling - Lawley Trace: } U = \sum_{i=1}^p \lambda_i = \text{trace} (\mathbf{W}^{-1}\mathbf{B})$$

$$\text{Roy's Largest Root: } \theta = \lambda_1$$

1-Way ANOVA – Multivariate Case - VI

Define: $p \equiv \#$ response variables ; $g \equiv \#$ of treatments (groups) ;
 $q \equiv df_{\text{Trts}} (g - 1 \text{ for } 1 - \text{Way AOV})$; $N \equiv \text{Total } \# \text{ of units}$

$$r = (N - g) - \frac{p - q + 1}{2} \quad u = \frac{pq - 2}{4} \quad t = \begin{cases} \sqrt{\frac{p^2 q^2 - 4}{p^2 + q^2 - 5}} & \text{if } p^2 + q^2 - 5 > 0 \\ 1 & \text{otherwise} \end{cases}$$

$$m_1 = \frac{N - g - p - 1}{2} \quad m_2 = \frac{|p - q| - 1}{2} \quad s_1 = \min(p, q) \quad s_2 = \max(p, q)$$

Wilks: $F_W = \left(\frac{1 - \Lambda^{1/t}}{\Lambda^{1/t}} \right) \left(\frac{rt - 2u}{pq} \right) \quad df_{1W} = pq \quad df_{2W} = rt - 2u$

Pillai: $F_P = \left(\frac{2m_1 + s_1 + 1}{2m_2 + s_1 + 1} \right) \left(\frac{V}{s_1 - V} \right) \quad df_{1P} = s_1 (2m_2 + s_1 + 1) \quad df_{2P} = s_1 (2m_1 + s_1 + 1)$

Hotelling Lawley: $F_{HL} = \frac{2(s_1 m_1 + 1)U}{s_1^2 (2m_2 + s_1 + 1)} \quad df_{1HL} = s_1 (2m_2 + s_1 + 1) \quad df_{2HL} = 2(s_1 m_1 + 1)$

Roy: $F_R = \frac{\theta (N - g - s_2 + q)}{s_2} \quad df_{1R} = s_2 \quad df_{2R} = N - g - s_2 + q$

Testing Equality of Covariance Matrices Among Treatments I

$$\Sigma_l = E \left\{ (\mathbf{X}_{lj} - \mu_l) (\mathbf{X}_{lj} - \mu_l)' \right\} \quad \mathbf{X}_{lj} = \begin{bmatrix} X_{lj1} \\ \vdots \\ X_{ljp} \end{bmatrix} \quad \mu_l = \begin{bmatrix} \mu_{l1} \\ \vdots \\ \mu_{lp} \end{bmatrix}$$

$$H_0 : \Sigma_1 = \dots = \Sigma_g = \Sigma$$

$$\text{Observed data} \rightarrow \bar{\mathbf{x}}_l = \frac{1}{n_l} \sum_{j=1}^{n_l} \mathbf{x}_{lj} \quad l = 1, \dots, g; j = 1, \dots, n_l$$

$$\mathbf{S}_l = \frac{1}{n_l - 1} \sum_{j=1}^{n_l} (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l) (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l)' \quad \mathbf{S}_{\text{Pooled}} = \frac{1}{N - g} \sum_{l=1}^g (n_l - 1) \mathbf{S}_l$$

Likelihood Ratio Test:

$$\Lambda = \prod_{l=1}^g \left(\frac{|\mathbf{S}_l|}{|\mathbf{S}_{\text{Pooled}}|} \right)^{\frac{n_l - 1}{2}}$$

$$M = -2 \ln \Lambda = -2 \sum_{l=1}^g \left[\left(\frac{n_l - 1}{2} \right) (\ln |\mathbf{S}_l| - \ln |\mathbf{S}_{\text{Pooled}}|) \right] = (N - g) \ln |\mathbf{S}_{\text{Pooled}}| - \sum_{l=1}^g (n_l - 1) \ln |\mathbf{S}_l|$$

Testing Equality of Covariance Matrices Among Treatments II

Box's Test (Works well when all $n_l \geq 20$, $p, g \leq 5$)

$$U = \left[\sum_{l=1}^g \frac{1}{n_l - 1} - \frac{1}{N - g} \right] \left[\frac{2p^2 + 3p - 1}{6(p + 1)(g - 1)} \right] \quad C = (1 - U) M$$

Under $H_0 : \Sigma_1 = \dots = \Sigma_g = \Sigma$:

$$C \sim \chi^2_{\nu} \quad \text{with} \quad \nu = \frac{1}{2}gp(p + 1) - \frac{1}{2}p(p + 1) = \frac{p(p + 1)(g - 1)}{2}$$

$$RR : C \geq \chi^2_{\nu}(\alpha) \quad p_{value} = P(\chi^2_{\nu} \geq C)$$

Plan

- 1 Paired Comparisons
 - Univariate Case
 - Multivariate Case
- 2 Comparing 2 Populations – Independent Samples
- 3 1-Way Analysis of Variance
 - Univariate Case
 - Multivariate Case
- 4 2-Way Analysis of Variance
 - Univariate (Balanced) Case
 - Multivariate (Balanced) Case - I
 - Profile Analysis

2-Way ANOVA – Univariate (Balanced) Case - I

2 Factors: 1 with g levels, other with b levels, n replicates per Treatment

$$X_{lkr} = \mu + \tau_l + \beta_k + \gamma_{lk} + \epsilon_{lkr} \quad l = 1, \dots, g; \quad k = 1, \dots, b; \quad r = 1, \dots, n$$

$$\epsilon_{lkr} \stackrel{i.i.d}{\sim} \mathcal{N}(0, \sigma^2) \quad \sum_{l=1}^g \tau_l = \sum_{k=1}^b \beta_k = \sum_{l=1}^g \gamma_{lk} = \sum_{k=1}^b \gamma_{lk} = 0$$

Observed data: $(x_{lkr} - \bar{x}) = (\bar{x}_l - \bar{x}) + (\bar{x}_k - \bar{x}) + (\bar{x}_{lk} - \bar{x}_l - \bar{x}_k + \bar{x}) + (x_{lkr} - \bar{x}_{lk})$

$$\begin{aligned} \sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (x_{lkr} - \bar{x})^2 &= bn \sum_{l=1}^g (\bar{x}_l - \bar{x})^2 + gn \sum_{k=1}^b (\bar{x}_k - \bar{x})^2 \\ &+ n \sum_{l=1}^g \sum_{k=1}^b (\bar{x}_{lk} - \bar{x}_l - \bar{x}_k + \bar{x})^2 + \sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (x_{lkr} - \bar{x}_{lk})^2 \end{aligned}$$

$$SS_{\text{TOTAL}} = SS_{\text{TRT1}} + SS_{\text{TRT2}} + SS_{\text{INT}} + SS_{\text{ERR}}$$

$$df_{\text{TOTAL}} = gbn - 1 \quad df_{\text{TRT1}} = g - 1 \quad df_{\text{TRT2}} = b - 1 \quad df_{\text{INT}} = (g - 1)(b - 1)$$

2-Way ANOVA – Univariate (Balanced) Case - II

$$H_0^\gamma : \gamma_{11} = \dots = \gamma_{gb} = 0 \quad TS : F_\gamma = \frac{MS_{INT}}{MS_{ERR}} = \frac{(SS_{INT}/((g-1)(b-1)))}{(SS_{ERR}/(gb(n-1)))}$$

$$RR : F_\gamma \geq F_{(g-1)(b-1), gb(n-1)}(\alpha) \quad pvalue = P(F_{(g-1)(b-1), gb(n-1)} \geq F_\gamma)$$

$$H_0^\tau : \tau_1 = \dots = \tau_g = 0 \quad TS : F_\tau = \frac{MS_{TRT1}}{MS_{ERR}} = \frac{(SS_{TRT1}/(g-1))}{(SS_{ERR}/(gb(n-1)))}$$

$$RR : F_\tau \geq F_{g-1, gb(n-1)}(\alpha) \quad pvalue = P(F_{g-1, gb(n-1)} \geq F_\tau)$$

$$H_0^\beta : \beta_1 = \dots = \beta_b = 0 \quad TS : F_\beta = \frac{MS_{TRT2}}{MS_{ERR}} = \frac{(SS_{TRT2}/(b-1))}{(SS_{ERR}/(gb(n-1)))}$$

$$RR : F_\beta \geq F_{b-1, gb(n-1)}(\alpha) \quad pvalue = P(F_{b-1, gb(n-1)} \geq F_\beta)$$

2-Way ANOVA – Multivariate (Balanced) Case - I

$$\mathbf{X}_{lkr} = \boldsymbol{\mu} + \boldsymbol{\tau}_l + \boldsymbol{\beta}_k + \boldsymbol{\gamma}_{lk} + \boldsymbol{\epsilon}_{lkr}$$

$$\begin{bmatrix} X_{lkr1} \\ \vdots \\ X_{lkrp} \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_p \end{bmatrix} + \begin{bmatrix} \tau_{l1} \\ \vdots \\ \tau_{lp} \end{bmatrix} + \begin{bmatrix} \beta_{k1} \\ \vdots \\ \beta_{kp} \end{bmatrix} + \begin{bmatrix} \gamma_{lk1} \\ \vdots \\ \gamma_{lkp} \end{bmatrix} + \begin{bmatrix} \epsilon_{lkr1} \\ \vdots \\ \epsilon_{lkrp} \end{bmatrix}$$

$$\epsilon_{lkr} \stackrel{i.i.d}{\sim} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}) \quad \text{and} \quad \sum_{l=1}^g \boldsymbol{\tau}_l = \sum_{k=1}^b \boldsymbol{\beta}_k = \sum_{l=1}^g \boldsymbol{\gamma}_{lk} = \sum_{k=1}^b \boldsymbol{\gamma}_{lk} = \mathbf{0}$$

Observed Data (Sums of Squares and Cross - Products Matrices):

$$\begin{aligned} \sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (\mathbf{x}_{lkr} - \bar{\mathbf{x}}) (\mathbf{x}_{lkr} - \bar{\mathbf{x}})' &= bn \sum_{l=1}^g (\bar{\mathbf{x}}_l - \bar{\mathbf{x}}) (\bar{\mathbf{x}}_l - \bar{\mathbf{x}})' + gn \sum_{k=1}^b (\bar{\mathbf{x}}_k - \bar{\mathbf{x}}) (\bar{\mathbf{x}}_k - \bar{\mathbf{x}})' \\ &+ n \sum_{l=1}^g \sum_{k=1}^b (\bar{\mathbf{x}}_{lk} - \bar{\mathbf{x}}_l - \bar{\mathbf{x}}_k + \bar{\mathbf{x}}) (\bar{\mathbf{x}}_{lk} - \bar{\mathbf{x}}_l - \bar{\mathbf{x}}_k + \bar{\mathbf{x}})' + \sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (\mathbf{x}_{lkr} - \bar{\mathbf{x}}_{lk}) (\mathbf{x}_{lkr} - \bar{\mathbf{x}}_{lk})' \end{aligned}$$

$$SSCP_{\text{TOTAL}} = SSCP_{\text{FACTOR1}} + SSCP_{\text{FACTOR2}} + SSCP_{\text{INT}} + SSCP_{\text{ERROR}}$$

2-Way ANOVA – Multivariate (Balanced) Case - II

MANOVA Table:

Source of Variation	Degrees of Freedom	SSCP Matrix
Factor 1	$g - 1$	$bn \sum_{l=1}^g (\bar{\mathbf{x}}_l - \bar{\mathbf{x}}) (\bar{\mathbf{x}}_l - \bar{\mathbf{x}})'$
Factor 2	$b - 1$	$gn \sum_{k=1}^b (\bar{\mathbf{x}}_k - \bar{\mathbf{x}}) (\bar{\mathbf{x}}_k - \bar{\mathbf{x}})'$
Interaction	$(g - 1)(b - 1)$	$n \sum_{l=1}^g \sum_{k=1}^b (\bar{\mathbf{x}}_{lk} - \bar{\mathbf{x}}_l - \bar{\mathbf{x}}_k + \bar{\mathbf{x}}) (\bar{\mathbf{x}}_{lk} - \bar{\mathbf{x}}_l - \bar{\mathbf{x}}_k + \bar{\mathbf{x}})'$
Error (Residual)	$gb(n - 1)$	$\sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (\mathbf{x}_{lkr} - \bar{\mathbf{x}}_{lk}) (\mathbf{x}_{lkr} - \bar{\mathbf{x}}_{lk})'$
Total (Corrected)	$gbn - 1$	$\sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (\mathbf{x}_{lkr} - \bar{\mathbf{x}}) (\mathbf{x}_{lkr} - \bar{\mathbf{x}})'$

2-Way ANOVA – Multivariate (Balanced) Case - III

Significance Tests for Model Parameters:

Interaction: $H_0^\gamma : \gamma_{11} = \dots = \gamma_{gb} = 0$

$$\Lambda_{\text{INT}}^* = \frac{|SSCP_{\text{ERROR}}|}{|SSCP_{\text{INT}} + SSCP_{\text{ERROR}}|} \quad RR : - \left[gb(n-1) - \frac{p+1-(g-1)(b-1)}{2} \right] \ln \Lambda_{\text{INT}}^* \geq \chi_{p(g-1)(b-1)}^2(\alpha)$$

Factor 1: $H_0^\tau : \tau_1 = \dots = \tau_g = 0$

$$\Lambda_{\text{FACTOR1}}^* = \frac{|SSCP_{\text{ERROR}}|}{|SSCP_{\text{FACTOR1}} + SSCP_{\text{ERROR}}|} \quad RR : - \left[gb(n-1) - \frac{p+1-(g-1)}{2} \right] \ln \Lambda_{\text{FACTOR1}}^* \geq \chi_{p(g-1)}^2(\alpha)$$

Factor 2: $H_0^\beta : \beta_1 = \dots = \beta_b = 0$

$$\Lambda_{\text{FACTOR2}}^* = \frac{|SSCP_{\text{ERROR}}|}{|SSCP_{\text{FACTOR2}} + SSCP_{\text{ERROR}}|} \quad RR : - \left[gb(n-1) - \frac{p+1-(b-1)}{2} \right] \ln \Lambda_{\text{FACTOR2}}^* \geq \chi_{p(b-1)}^2(\alpha)$$

2-Way Analysis of Variance – Multivariate Case - IV

4 Approximate F -Tests Based on Eigenvalues of $SSCP_{Error}^{-1}SSCP_{Factor}$:

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p > 0$$

Wilks' Λ : $\Lambda_{Factor} = \prod_{i=1}^p \frac{1}{1 + \lambda_i} = \frac{|SSCP_{Error}|}{|SSCP_{Factor} + SSCP_{Error}|}$

Pillai's Trace: $V_{Factor} = \sum_{i=1}^p \frac{\lambda_i}{1 + \lambda_i} = \text{trace} (SSCP_{Factor} (SSCP_{Factor} + SSCP_{Error})^{-1})$

Hotelling - Lawley Trace: $U_{Factor} = \sum_{i=1}^p \lambda_i = \text{trace} (SSCP_{Error}^{-1}SSCP_{Factor})$

Roy's Largest Root: $\theta_{Factor} = \lambda_1$

2-Way Analysis of Variance – Multivariate Case - V

Define: $p \equiv \#$ response variables, $g \equiv \#$ of levels of Trt1, $b \equiv \#$ of levels of Trt2,
 $q \equiv df_{\text{Factor}}$ ($g - 1$ for Trt1, $b - 1$ for Trt2, $(g - 1)(b - 1)$ for Int)
 $N \equiv$ Total $\#$ of units, $r = (N - gb) - \frac{p-q+1}{2}$, $u = \frac{pq-2}{4}$,

$$t = \begin{cases} \sqrt{\frac{p^2q^2-4}{p^2+q^2-5}} & \text{if } p^2 + q^2 - 5 > 0 \\ 1 & \text{otherwise} \end{cases}$$

$$m_1 = \frac{N - gb - p - 1}{2} \quad m_2 = \frac{|p - q| - 1}{2} \quad s_1 = \min(p, q) \quad s_2 = \max(p, q)$$

Wilks: $F_W = \left(\frac{1 - \Lambda^{1/t}}{\Lambda^{1/t}} \right) \left(\frac{rt - 2u}{pq} \right) \quad df_{1W} = pq \quad df_{2W} = rt - 2u$

Pillai: $F_P = \left(\frac{2m_1 + s_1 + 1}{2m_2 + s_1 + 1} \right) \left(\frac{V}{s_1 - V} \right) \quad df_{1P} = s_1(2m_2 + s_1 + 1) \quad df_{2P} = s_1(2m_1 + s_1 + 1)$

Hotelling Lawley: $F_{HL} = \frac{2(s_1m_1 + 1)U}{s_1^2(2m_2 + s_1 + 1)} \quad df_{1HL} = s_1(2m_2 + s_1 + 1) \quad df_{2HL} = 2(s_1m_1 + 1)$

Roy: $F_R = \frac{\theta(N - gb - s_2 + q)}{s_2} \quad df_{1R} = s_2 \quad df_{2R} = N - gb - s_2 + q$

2-Way ANOVA – Multivariate (Balanced) Case - VI

Bonferroni Simultaneous Confidence Intervals for Comparing Main Effects:

Factor 1: $\tau_{lj} - \tau_{mi}$ Number of Comparisons: $C_1 = p \binom{g}{2} = \frac{pg(g-1)}{2}$

$$(\bar{x}_{lj} - \bar{x}_{mi}) \pm t_{gb(n-1)} \left(\frac{\alpha}{2C_1} \right) \sqrt{\frac{E_{ij}}{gb(n-1)} \left(\frac{2}{bn} \right)}$$

with $E_{ij} \equiv i^{th}$ diagonal element of $SSCP_{\text{ERROR}}$

Factor 2: $\beta_{ki} - \beta_{qi}$ Number of Comparisons: $C_2 = p \binom{b}{2} = \frac{pb(b-1)}{2}$

$$(\bar{x}_{ki} - \bar{x}_{qi}) \pm t_{gb(n-1)} \left(\frac{\alpha}{2C_2} \right) \sqrt{\frac{E_{ij}}{gb(n-1)} \left(\frac{2}{gn} \right)}$$

with $E_{ij} \equiv i^{th}$ diagonal element of $SSCP_{\text{ERROR}}$

Profile Analysis - I

- Two or More Groups to Be Compared
- Measurements made on p variables (of similar scales) on each experimental/sampling unit
- Typical Tests:
 - 1 Parallel Profiles Among Groups (For 2 Groups:
 $H_{0i} : \mu_{1i} - \mu_{11} = \mu_{2i} - \mu_{21} \quad i = 2, \dots, p$)
 - 2 Assuming Parallel Profiles, are Profiles Coincident (For 2 Groups: $H_{0i} : \mu_{1i} = \mu_{2i} \quad i = 1, \dots, p$)
 - 3 Assuming Coincident Profiles, Level Profiles (For 2 Groups:
 $H_0 : \mu_{11} = \mu_{21} = \dots = \mu_{1p} = \mu_{2p}$)

1) Parallel Profiles: $H_{01i} : \mu_{1i} - \mu_{11} = \mu_{2i} - \mu_{21} \quad i = 2, \dots, p$

$$TS : T_1^2 = [\mathbf{C}_1 (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)]' \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{C}_1 \mathbf{S}_{\text{Pooled}} \mathbf{C}_1' \right]^{-1} [\mathbf{C}_1 (\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)]$$

$$RR : T_1^2 \geq \left[\frac{(n_1 + n_2 - 2)(p - 1)}{n_1 + n_2 - p} \right] F_{p-1, n_1+n_2-p}(\alpha)$$

2) Assuming Parallel, Coincident Profiles:

$$H_{02i} : \mu_{1i} = \mu_{2i} \quad i = 1, \dots, p \quad H_{02} : \mathbf{1}'\mu_1 = \mathbf{1}'\mu_2 \Rightarrow \mathbf{1}'(\mu_1 - \mu_2) = \mathbf{0}$$

$$TS : T_2^2 = [\mathbf{1}'(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)]' \left[\left(\frac{1}{n_1} + \frac{1}{n_2} \right) \mathbf{1}' \mathbf{S}_{\text{Pooled}} \mathbf{1} \right]^{-1} [\mathbf{1}'(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)], \quad RR : T_2^2 \geq F_{1, n_1+n_2-2}(\alpha)$$

3) Assuming, Coincident, Level Profiles:

$$H_{03} : \mu_{11} = \mu_{21} = \dots = \mu_{1p} = \mu_{2p} \quad H_{03} : \mathbf{C}_1 \mu = \mathbf{0} \quad \mu = \frac{1}{2}(\mu_1 + \mu_2)$$

$$\bar{\mathbf{x}} = \frac{1}{n_1 + n_2} \left[\sum_{j=1}^{n_1} \mathbf{x}_{1j} + \sum_{j=1}^{n_2} \mathbf{x}_{2j} \right] = \frac{n_1 \bar{\mathbf{x}}_1 + n_2 \bar{\mathbf{x}}_2}{n_1 + n_2}$$

$$TS : T_3^2 = (n_1 + n_2) [\mathbf{C}_1 \bar{\mathbf{x}}]' [\mathbf{C}_1 \mathbf{S}_{\text{Pooled}} \mathbf{C}_1']^{-1} [\mathbf{C}_1 \bar{\mathbf{x}}]$$

$$RR : T_3^2 \geq \left[\frac{(n_1 + n_2 - 1)(p - 1)}{n_1 + n_2 - p + 1} \right] F_{p-1, n_1+n_2-p+1}(\alpha)$$

Repeated Measures Designs

- Each Subject/Unit receives each Treatment – Randomized Block Design
 - Measurements within subjects are correlated due to random subject effects
 - Can be used to get more precise treatment effects when subject-to-subject variability is large
 - Can have problems with carryover or learning when subject repeats same task under different treatments/Conditions
- Subjects/Units are assigned to only one Treatment and Observed at Multiple Points in Time – Completely Randomized Design
 - Measurements within subjects are correlated due to random subject effects
 - Can Estimate Time Effects and Treatment/Time Interactions
- Growth Curves – Individual Subjects observed over time and growth within Subjects Modeled with a Linear or Nonlinear Regression Model

Growth Curves – Pothoff and Roy Model - I

1) Single Treatment - n Subjects Observed at time points $t_1, \dots, t_p$

Quadratic Model:

$$\mathbf{X}_j = \begin{bmatrix} X_{j1} \\ \vdots \\ X_{jp} \end{bmatrix} \quad E \{ \mathbf{X}_j \} = \begin{bmatrix} \beta_0 + \beta_1 t_1 + \beta_2 t_1^2 \\ \vdots \\ \beta_0 + \beta_1 t_p + \beta_2 t_p^2 \end{bmatrix} = \begin{bmatrix} 1 & t_1 & t_1^2 \\ \vdots & \vdots & \vdots \\ 1 & t_p & t_p^2 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} = \mathbf{B}\beta$$

$$V \{ \mathbf{X}_j \} = \Sigma, \quad j = 1, \dots, n.$$

2) g Treatment Groups – n_l Subjects in Group l: Quadratic Model:

$$\mathbf{X}_{lj} = \begin{bmatrix} X_{lj1} \\ \vdots \\ X_{lj p} \end{bmatrix} \quad V \{ \mathbf{X}_{lj} \} = \Sigma \quad l = 1, \dots, g; \quad j = 1, \dots, n_l$$

$$E \{ \mathbf{X}_{lj} \} = \begin{bmatrix} \beta_{l0} + \beta_{l1} t_1 + \beta_{l2} t_1^2 \\ \vdots \\ \beta_{l0} + \beta_{l1} t_p + \beta_{l2} t_p^2 \end{bmatrix} = \begin{bmatrix} 1 & t_1 & t_1^2 \\ \vdots & \vdots & \vdots \\ 1 & t_p & t_p^2 \end{bmatrix} \begin{bmatrix} \beta_{l0} \\ \beta_{l1} \\ \beta_{l2} \end{bmatrix} = \mathbf{B}_l \beta_l$$

$$q \text{ - order polynomial model: } \mathbf{B} = \begin{bmatrix} 1 & t_1 & \dots & t_1^q \\ \vdots & \vdots & \ddots & \vdots \\ 1 & t_p & \dots & t_p^q \end{bmatrix} \quad \beta_l = \begin{bmatrix} \beta_{l0} \\ \beta_{l1} \\ \vdots \\ \beta_{lp} \end{bmatrix}$$

Growth Curves – Pothoff and Roy Model - II

$$\text{Observed data: } \mathbf{x}_{lj} = \begin{bmatrix} x_{lj1} \\ \vdots \\ x_{ljg} \end{bmatrix} \quad l = 1, \dots, g; \quad j = 1, \dots, n_l \quad \rightarrow \quad \bar{\mathbf{x}}_l = \frac{1}{n_l} \sum_{j=1}^{n_l} \mathbf{x}_{lj}$$

$$\mathbf{S}_l = \frac{1}{n_l - 1} \sum_{j=1}^{n_l} (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l) (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l)' \quad \mathbf{S}_{\text{Pooled}} = \frac{1}{N - g} \sum_{l=1}^g (n_l - 1) \mathbf{S}_l = \frac{1}{N - g} \mathbf{W}$$

Maximum Likelihood/Estimated Generalized Least Squares Estimator (q^{th} – order polynomial model):

$$\hat{\beta}_l = (\mathbf{B}' \mathbf{S}_{\text{Pooled}}^{-1} \mathbf{B})^{-1} \mathbf{B}' \mathbf{S}_{\text{Pooled}}^{-1} \bar{\mathbf{x}}_l \quad l = 1, \dots, g$$

$$\text{cov} \{ \hat{\beta}_l \} = \frac{K}{n_l} (\mathbf{B}' \mathbf{S}_{\text{Pooled}}^{-1} \mathbf{B})^{-1} \quad K = \frac{(N - g)(N - g - 1)}{(N - g - p + q)(N - g - p + q + 1)} \quad \text{cov} \{ \hat{\beta}_l, \hat{\beta}_h \} = \mathbf{0} \quad l \neq h$$

H_0 : q^{th} - order polynomial model is adequate

$\mathbf{W} \equiv$ within groups Sum of Squares and Cross - Products Matrix:

$$\mathbf{W} = \text{SSCP}_{\text{ERROR}} = \sum_{l=1}^g (n_l - 1) \mathbf{S}_l \text{ with } \# \text{ of estimated parameters} = gp$$

$$\mathbf{W}_q = \sum_{l=1}^g \sum_{j=1}^{n_l} (\mathbf{x}_{lj} - \mathbf{B} \hat{\beta}_l) (\mathbf{x}_{lj} - \mathbf{B} \hat{\beta}_l)' \text{ with } \# \text{ of estimated parameters} = g(q + 1)$$

$$\text{Wilks' } \Lambda^* = \frac{|\mathbf{W}|}{|\mathbf{W}_q|} \quad RR : - \left(N - \frac{1}{2}(p - q + g) \right) \ln \Lambda^* \geq \chi_{g(p-q-1)}^2(\alpha)$$